Periods & Amplitudes

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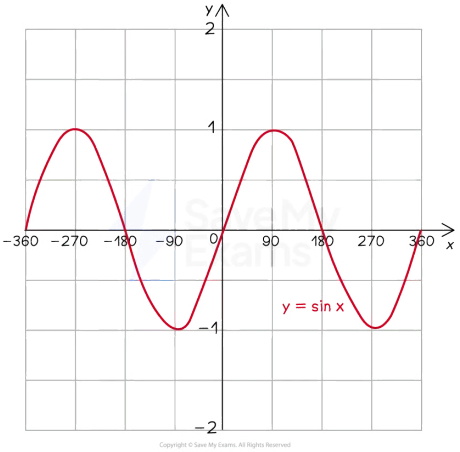
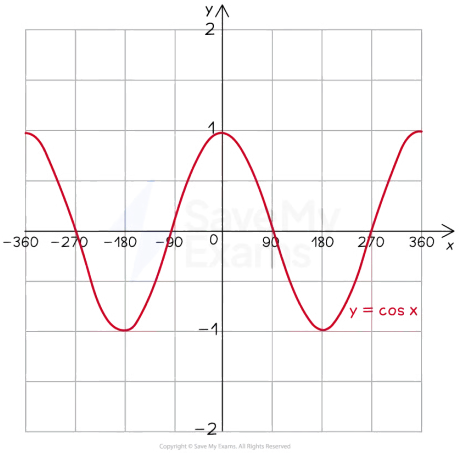
Periods & Amplitudes

What are oscillating graphs?

- The **sine** and **cosine** graphs are oscillating graphs
  - Both graphs oscillate between **a maximum and a minimum point**
  - A cycle is completed when the graph returns an initial starting point
  - The cycle is **repeated indefinitely** in both the positive and negative *x* directions





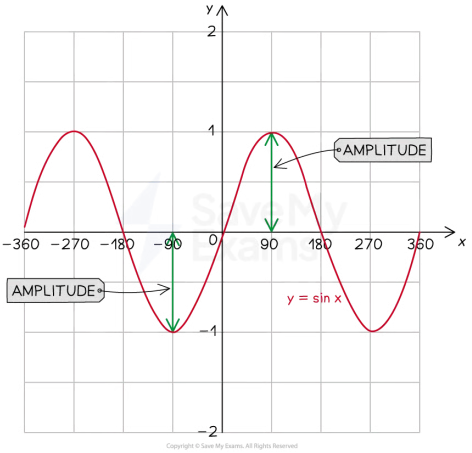
Help me understand

Stuck on something?

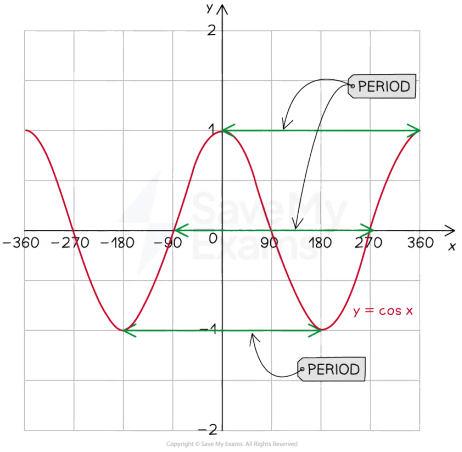
Ask questions and get help with exactly what you're revising.

Got it

- An oscillating graph has the following key features:
  - The **amplitude** is half the vertical distance from the maximum point to the minimum point
    - This is the distance from the horizontal line of symmetry (often the x-axis) to the minimum or maximum point
  - E.g. For the graph $y = \sin x$ the amplitude is 1



- The **period** is the distance along the x-axis to complete a full oscillation
  - Maximum point to maximum point
  - Minimum point to minimum point
  - x-intercept to next but one x-intercept
  - E.g. For the graph $y = \cos x$ the period is 360°



How do I draw an oscillating graph from its equation?

- An oscillating function has **equation**
  - $f(x) = a \sin(bx)$
  - $f(x) = a \cos(bx)$
- The a represents the **amplitude** of the function
  - The bigger the value of a the bigger the **range** of values of the function

symmetry) to the maximum/minimum point would be 5 units

- E.g. For the graph $y = 2 + \frac{1}{5}\cos x$
  - The amplitude is $\frac{1}{5}$ of a unit
  - The vertical distance from the line $y = 2$ to the maximum/minimum point would be $\frac{1}{5}$ of a unit

- The b determines the **period** of the function
  - The bigger the value of b the **quicker** the function repeats a cycle
  - The period is $\frac{360°}{b}$
    - A **larger value** of b produces a **shorter period**
  - E.g. For the graph $y = \sin(2x)$
    - The period is $\frac{360}{2} = 180°$
    - The graph has been 'squashed' by a factor of 2 horizontally
    - Two complete cycles will occur in an interval of $360°$
  - E.g. For the graph $y = \sin\left(\frac{1}{2}x\right)$
    - The period is $\frac{360}{\frac{1}{2}} = 360 \times \frac{2}{1} = 720°$
    - The graph has been 'stretched'
    - Half a cycle will occur in an interval of $360°$

Examiner Tips and Tricks

- You can use your **graphic display calculator** to help you interpret or compare trigonometric graphs

Worked Example

The diagram below shows the graph with equation $y = a\sin(bx)$, where a and b are integer values.

(a) Write down the period.

The graph intersects the x axis at 0, then again at 45° and again at 90°

One full cycle is complete over an interval of 90°

Period = 90°

(b) Write down the amplitude.

The graph goes to a maximum of 3 and a minimum of -3

Amplitude = 3

(c) Write down the values of a and b.

a is the amplitude

The period is $\frac{360°}{b}$

$$90 = \frac{360}{b}$$

$$b = \frac{360}{90}$$

$$b = 4$$

$$a = 3$$
$$b = 4$$

(d) Give a reason as to why the picture cannot be modelled by $y = a\cos(bx)$.

The curve in the diagram goes through the origin $(0, 0)$ but a curve with equation of the form $y = a\cos(bx)$ would intercept the y-axis at $(0, a)$

Any suitable reason as to how a cosine graph is different from a sine graph

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